ALISA OMELCHENKO

347-570-5722 | alisa.a.omelchenko@gmail.com | linkedin.com/in/alisa-omelchenko | Github @AlisaOmel
AlisaOmel.github.io | Pittsburgh, PA, 15207

RESEARCH PROFILE

Creative fourth-year PhD candidate in the CMU-Pitt joint Computational Biology program focused on problem-solving. Passionate about collaboratively developing accessible tools that empower biologists and move science forward. With hands-on experience across both wet-lab and computational research, I bridge the gap between experimental biology and computer science. My work spans interaction-based language models, multi-omic network analysis, and protein modeling with applications in immunology and disease. I am especially committed to keeping computational approaches grounded in biological relevance, ensuring the methods I build stay focused on answering real questions. Additionally, I prioritize communication through mentoring future generations and advocating for science and education policy.

EDUCATION

CMU-Pitt Joint Computational Biology: Pittsburgh, Pennsylvania *Sept 2022- Present*
PhD Candidate in Computational Biology

NYU Tandon School of Engineering: Brooklyn, New York *Graduated May 2018*
M.S. Biotechnology *GPA: 3.94*
B.S. BioMolecular Sciences (Magna Cum Laude) *GPA: 3.61*

SKILLS

Computer Languages: Python, R, Bash
Computational Skills: HPC (Slurm, LSF), deep learning (CNNs, transformers), protein and interaction language models (LLMs, pLMs/iLMs), interpretable ML, multi-omic network analysis, protein modeling and visualization (AlphaFold3, OpenMM, PyMOL, RDKit), single-cell analysis (scRNA-seq, Multiome), ImageJ/FIJI, fMRI analysis and data visualization.
Laboratory Skills: Genotyping/PCR, ELISA, flow cytometry, calcium imaging, confocal microscopy, molecular cloning, animal handling (mice, *Drosophila*)

RESEARCH EXPERIENCE

St. Jude Center for Applied Bioinformatics: Summer Research Intern *June 2026-Present*
Advised by: Gang Wu and Jeremy Chase Crawford
- Worked with the genomics team to predict patient-specific neoepitopes from a B-ALL cohort and identified patterns associated with relapse and B-ALL subtypes.
- Generated a reproducible and scalable pipeline that can be easily applied to new datasets.
- Mentored 2 high school research immersion (HSRI) interns and guided them through hands-on research.

CMU-Pitt Joint Computational Biology: Graduate Student *Sept 2022-Present*
Advised by: Jishnu Das
- Developed SWING, a first-in-class interaction-based vocabulary language model (iLM) to predict peptide:peptide interactions and perturbations collaboratively (Published in Nat. Methods).
- Integrated multi-omic datasets and performed network-based analysis to identify a core T follicular helper (Tfh) cell network that is consistent across contexts (Accepted in Principle at Nat. Commun.).
- Collaborated to identify novel mechanisms of HIV infection in Tfh cells (In Revision at Nat. Commun.).
- Extended iLMs to capture ternary (pMHC:TCR) and non-peptide partners (small-molecule:protein) interactions, enabling generalizable, patient-specific predictions for clinical translation.

Virginia Tech School of Neuroscience Ni Lab: Laboratory Manager *July 2019-July 2022*
Advised by: Lina Ni
- Created an ImageJ plugin to automate data extraction and processing of experiments (Published in JoVE).
- Identified the function, molecular components, and properties in the thermosensory pathway of *Drosophila* larvae (Published in JEB & Front. Mol. Neurosci.).
- Designed automated experimental workflows for fly and larval tracking to ensure reproducible results and speed up data analysis (Published in JEB).

• Generated new fly strains for analysis of gene function through molecular cloning and recombination.

Landos Biopharma: Laboratory Technician July *2018- July 2019*
Advised by: *Josep Bassaganya-Riera*
• Worked on target identification and mechanism-of-action studies by evaluating differentially expressed gene patterns.
• Collected and analyzed data from necropsies, cell cultures, flow cytometry, ELISA, and western blots.
• Handled, monitored, and performed routine medical treatments on mice, rats, and pigs.

NYU Langone Health: Volunteer Researcher *April 2018-June 2018*
Advised by: Timothy J. Cardozo
• Designed structure and evaluated stability of antibody eliciting epitopes for staph database and staph vaccines.
• Analyzed alignments to predict protein structures through homology modeling of proteins not in the PDB.

Nathan Kline Institute: Research Assistant *September 2016- May 2018*
Advised by: Cameron R. Craddock
• Determined functional similarities of the brain between healthy individuals with comparable personality traits.
• Mapped cluster differences of fMRI data on brain templates to visualize subgroup differences in activation.
• Assessed the quality of 225 skull-stripped brain images and manually edited them.

Child Mind Institute: Volunteer Research Assistant *September 2016-May 2018*
Advised by: Steven Giavasis
• Improved the speed, usability, and statistical implementation of Python scripts used by the institute.
• Enhanced a Python script to utilize a Docker container and parallelize the data processing increasing the speed by 30%. The script is included in the current version of C-PAC software.

PUBLICATIONS

Omelchenko, A. A., Rahman, S. A., Viswanadham, V. V., Yuen, G. J., Estrada, P. M. D. R., D'Onofrio, V., ... & Das, J. (2025). A unified network systems approach uncovers a core novel program underlying T follicular helper cell differentiation. *bioRxiv*. [Accepted in Principle at Nature Communications]

Ribeiro, S.P.*, Del Rio Estrada, P. M.*, Georgakis, S., Orfanakis, M., **Omelchenko, A.**, Coirada, F., Dos Santos, J., et al. (2026). IL-10-and TGF-β-Driven BCL6 Expression Suppresses Antiviral Defenses and Renders Lymph Node T Follicular Helper Cells Permissive to HIV Infection. *Preprint*. [In Revision at Nature Communications]

Omelchenko, A. A.*, Siwek, J. C.*, Chhibbar, P.*, Arshad, S., Rosengart, A., Nazarali, I., ... & Das, J. (2025). Sliding Window Interaction Grammar (SWING): a generalized interaction language model for peptide and protein interactions. *Nature Methods*, 1-13.

Omelchenko, A. A., Bai, H., Hussain, S., Tyrrell, J. J., Klein, M., & Ni, L. (2022). TACI: An ImageJ Plugin for 3D Calcium Imaging Analysis. JoVE (Journal of Visualized Experiments), (190), e64953.

Omelchenko, A. A., Bai, H., Spina, E. C., Tyrrell, J. J., Wilbourne, J. T., & Ni, L. (2022). Cool and warm ionotropic receptors control multiple thermotaxes in Drosophila larvae. *Frontiers in Molecular Neuroscience*.
Huda, A.*, **Omelchenko, A. A.***, Vaden, T. J.*, Castaneda, A. N., & Ni, L. (2022). Responses of different Drosophila species to temperature changes. *Journal of Experimental Biology*, *225*(11), jeb243708.

Tyrrell, J. J., Wilbourne, J. T., **Omelchenko, A. A.**, Yoon, J., & Ni, L. (2021). Ionotropic Receptor-dependent cool cells control the transition of temperature preference in Drosophila larvae. *PLoS genetics*, *17*(4), e1009499. https://doi.org/10.1371/journal.pgen.1009499

Wang, X., Li, X. H., Cho, J. W., Russ, B. E., Rajamani, N., **Omelchenko, A.**, Ai, L., Korchmaros, A., Sawiak, S., Benn, A.R., Garcia-Saldivar, P., Wang, Z., Kalin N.H., Schroeder, C.E., Craddock, R.C., Fox, A.S., Evans, A.C., Messinger, A., Milham, M.P, Xu, T. (2021). U-Net Model for Brain Extraction: Trained on Humans for Transfer to Non-human Primates. *NeuroImage*, 118001. https://doi.org/10.1016/j.neuroimage.2021.118001

*These authors contributed equally

CONFERENCES and AWARDS

Presentations:

A unified network systems approach uncovers a core novel program underlying Tfh differentiation. Oral Presentation at: AAI, April 2026.

Computational approaches to understand immune system modulation through protein interactions. Sip of Science, December 2025.

A unified network systems approach uncovers a core novel program underlying Tfh differentiation. Oral Presentation at: Center for Systems Immunology Retreat, November 2025

A unified network systems approach uncovers a core novel program underlying Tfh differentiation. Oral Presentation at: ISMB/ECCB, July 2025.

Sliding Window INteraction Grammar (SWING): a generalized interaction language model for peptide and protein interactions, Oral Presentation at: Graduate Research Symposium, November 2024

Sliding Window INteraction Grammar (SWING): a generalized interaction language model for peptide and protein interactions. Oral Presentation at: ISCB GLBio, May 2024

SWING- A generalizable language model for protein and peptide interactions, Oral Presentation at: Center for Systems Immunology Retreat, September 2023

Ionotropic Receptor-dependent warm cells in *Drosophila* larvae, Oral Presentation at: VT School of Neuroscience: Summer Research Retreat, August 2020

Posters:

Omelchenko, A. A., Rahman, S. A., Viswanadham, V. V., Yuen, G. J., Estrada, P. M. D. R., D'Onofrio, V., ... & Das, J. (2025). A unified network systems approach uncovers a core novel program underlying T follicular helper cell differentiation. Poster presented at: 25th Annual Immunology Retreat, October 2025.

Omelchenko, A. A., Rahman, S. A., Viswanadham, V. V., Yuen, G. J., Estrada, P. M. D. R., D'Onofrio, V., ... & Das, J. (2025). A unified network systems approach uncovers a core novel program underlying T follicular helper cell differentiation. Poster presented at: 23rd Annual Immunology Retreat, October 2023.

Omelchenko A. A., Siwek, J., Chhibbar, P., Rosengart A., Koes D., Joglekar A., Das, J., A generalized language model for predicting perturbations of protein-protein and MHC:Peptide interactions. Poster presented at: Graduate Research Symposium, November 2023.

Omelchenko A. A., Siwek, J., Chhibbar, P., Rosengart A., Koes D., Joglekar A., Das, J., A generalized language model for predicting perturbations of protein-protein and MHC:Peptide interactions. Poster presented at: 21st Annual Immunology Retreat, October 2023.

Omelchenko, A. A., Tyrrell, J. J., Wilbourne, J. T., & Ni, L. Ionotropic Receptor-dependent dorsal organ warm cells mediate warm sensing in Drosophila larvae. Poster presented at: SfN, November 2021; Virtual

Tyrrell, J. J., Wilbourne, J. T., Omelchenko, A. A., Yoon, J., & Ni, L. Ionotropic Receptor-dependent cool cells control the transition of temperature preference in flies. Poster presented at: VT Molecular and Cellular Biology Summer Event, August 2021; Blacksburg, Virginia.

Honors and Awards:

Travel Awards:
- St. Jude Travel Award for BioHackathon participation
- BGSA Travel Award for Abstract at ISMB 2025 Conference and Talk
- CRA-WP Grad Cohort For Women Workshop Travel Award 2023
- Brainhack LA Travel Award 2016

Presentation Awards:
- AAI Trainee Abstract Award- AAI 2026
- Most Technically Impressive Project – St. Jude Hackathon 2025
- Best Abstract – Center for Systems Immunology 2025
- Best Abstract – Graduate Research Symposium 2024
- Best Talk – CPCB Retreat 2024
- Best Abstract – Center for Systems Immunology 2023

Academic Awards and Scholarships:
- Dean's List- Awarded for academic achievement 3 years in a row 2015-2018
- Trio Scholar—2013-2018
- STEM Women in Engineering Scholarship—2013-2018

Leadership and Service

St. Jude HSRI Mentor – Mentored 2 high school interns in the Research Immersion Program. June 2026-Present

Pennsylvania Woman Work Mentor – Mentoring individuals who are ready for their next career step and need guidance entering the workforce. July 2025-Present

Allegheny County Science Policy Member- Engage in science policy, advocacy, and communication at a local, state, and national level. May 2026-Present

BGSA CPCB Program Representative – Represent the CPCB graduate program in the School of Medicine student government. September 2023-Present

CPCB Steering Committee- Graduate representative from the CPCB program chosen to meet with the directors across CMU and PITT to steer the progress of the program. September 2024-September 2025

CPCB GSA Liaison – Connected the CPCB graduate program to the larger network of the School of Medicine student government. September 2023-July 2025

TechBIO REU mentor – Advised undergraduate students in research ethics as part of the TechBIO REU program. June 2023-August 2023, June 2024-August 2024, June 2025-July 2025

Book Club Mentor - Guided first-year students and facilitated their transition into the graduate program through discussions of assigned readings. September 2023-May 2025

First-Year Ambassador Mentor - Provided assistance and support to incoming students to assist in their acclimation to the graduate school. September 2023-2024

TRAINING

Teaching:

Teaching Assistant. Computational Structural Biology, University of Pittsburgh, *Sept 2023-Dec 2023*

- Developed curriculum, lesson plans, and course materials and led weekly recitations.
- Graded assignments and final projects and held weekly office hours.

Guest Lectures and Invited Talks:

"A Brief Overview of Computational Structural Biology," guest lecture, Data to Knowledge course, University of Pittsburgh, June 2023.

"How to Network at GLBio," invited talk, ISCB GLBio Workshop for Early Career Researchers, May 2024.

Relevant Graduate Coursework:

- Machine Learning for Biomedical Applications (PITT)
- 10-701 Introduction to Machine Learning (CMU)
- Genomics (CMU)
- Cellular and Systems Modeling (PITT)
- Computational Structural Biology (PITT)
- Essential Mathematics and Statistics for Scientists (CMU)
- Computation for Data Science I (Virginia Tech)
- Problem Solving in Genetics, Bioinformatics, and Computational Biology (Virginia Tech)
- Computer-Aided Drug design (NYU)
- Protein Engineering (NYU)
- Biocatalysts (NYU)
- Biosensors and Biochips (NYU)
- Immunology (NYU)